The S&P 500 Variance Risk Premium: Gradient-Boosting Forecasts and the Economic Value of the Signal

Revised and extended version of a Master's thesis

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Abstract

This paper asks whether a gradient-boosting forecast of the S&P 500 variance risk premium (VRP) improves a delta-hedged short-volatility strategy, net of costs and out of sample. Using monthly SPX options from 2018 to 2025, the premium is persistent: implied volatility exceeded subsequently realised volatility on 87.4% of sessions. An XGBoost model with 22 predictors and purged cross-validation lowers the mean squared error by 6% relative to the historical mean, but the improvement is not significant under the Diebold–Mariano test. At the trading layer the signal adds no value. Every out-of-sample forecast is positive, so the model-driven strategy coincides with unconditional selling, and the forecast does not rank trade outcomes. The advantage over a naive rule reported in the original thesis does not come from the model; the comparison was not like-for-like. The economic outcome depends mainly on hedge calibration and on how collateral is treated. With interest-bearing collateral, sales at the bid and the original hedge band, systematic selling earns a Sharpe ratio of 0.23, but that band was chosen in sample and its optimum is not stable out of sample. A synthetic variance swap confirms these conclusions without any execution assumptions and puts the February 2018 loss at about twenty months of average premium. No difference between strategies is statistically significant.

Keywords: variance risk premium, implied volatility, gradient boosting, purged cross-validation, delta hedging, variance swaps.
JEL classification: C53, G13, G17.

About this version. This document revises the Master's thesis *Arbitraje estadístico de volatilidad: modelado del Variance Risk Premium mediante Machine Learning* (Statistical volatility arbitrage: modelling the variance risk premium with machine learning), MSc in Financial Planning and Advice (Máster en Asesoramiento y Planificación Financiera), Universidad Rey Juan Carlos, July 2026. It corrects the interpretation of the comparison between the forecast-driven strategy and the systematic rule (Section 5.4) and adds nested validation, naive benchmarks, statistical tests, execution costs, collateral interest and an analysis with synthetic variance swaps (Sections 5.2 to 5.9). Code and results are reproducible at `https://github.com/borja-izquierdo/vrp-sp500-options`.

1 Introduction

The volatility implied by index option prices tends to exceed the volatility the index subsequently realises. This gap, the variance risk premium (VRP), is one of the best-documented regularities in derivatives markets (Carr and Wu, 2009), and it rewards those who sell volatility. Harvesting it, however, carries a highly asymmetric risk profile: small, frequent gains against rare but very large losses. On 5 February 2018 the VIX went from 18.44 to 37.32 in a single session, and inverse VIX exchange-traded products lost about 97% of their value before the next open (Augustin

et al., 2021). On 21 February Credit Suisse announced the early redemption of XIV at USD 5.99 per note (Varadarajan and Schenk-Hoppé, 2023).

The question of this paper is whether a machine-learning forecast of the premium allows it to be sold more efficiently than a mechanical rule; that is, whether it improves the risk–return profile of a delta-hedged short-volatility strategy, net of costs and out of sample. To answer it I build a complete system: download and cleaning of SPX option chains, 22 predictors, XGBoost and LightGBM models with a Huber loss and purged cross-validation, and an event-driven backtest engine that sells SPX straddles and delta-hedges them with commissions and slippage.

The contributions are four. First, the system is reproducible end to end with data any researcher can purchase. Second, forecast accuracy is assessed against naive benchmarks with the Diebold and Mariano (1995) test, not only in absolute terms. Third, forecast quality is separated from economic value: I measure whether the forecast ranks trade outcomes, whether its magnitude improves entry or position sizing, and how much of the result comes from costs, collateral and hedging. Finally, the premium is also measured with synthetic variance swaps, which covers the whole period (including 2021, for which no option chains are available) and quantifies the February 2018 loss.

The results are mixed and are reported without defensive hedging:

- The premium exists and is persistent: positive on 87.4% of sessions, with a mean of 4.95 volatility points.
- The model forecasts better than the naive benchmarks on average, but not in every period, and the improvement in squared error is not significant.
- The signal does not reach the trading layer. Every forecast of the selected model is positive, so the forecast-driven strategy is equivalent to always selling. Its rank correlation with the outcome of each trade is 0.05.
- The comparison with the systematic rule in the original thesis was not like-for-like: the rule's trade count included a period before the evaluation window. On a like-for-like basis there is no advantage attributable to the model.
- What drives the result is risk management: the width of the hedge band and, to a lesser extent, interest on collateral. None of the differences between strategies is statistically significant with 6.7 years and 64 trades.

The rest of the paper is organised as follows. Section 2 summarises the conceptual framework. Section 3 describes the data and Section 4 the methodology. Section 5 presents the results, Section 6 discusses limitations and extensions, and Section 7 concludes.

2 Conceptual framework

2.1 The variance risk premium

A variance swap is a forward contract on realised variance. At maturity the buyer receives $N_{\text{var}}\,(RV_{t,T} - K^2)$, where $RV_{t,T}$ is the annualised variance of daily returns between t and T and K^2 is the agreed strike. Because entry is costless, K^2 is the expectation of realised variance under the risk-neutral measure $\mathbb{Q}$, and Carr and Wu (2009) show that it can be replicated with a static portfolio of out-of-the-money options. The variance risk premium is the difference between the risk-neutral and the physical ($\mathbb{P}$) expectation:

$$\text{VRP}_t = \mathbb{E}_t^{\mathbb{Q}}[RV_{t,T}] - \mathbb{E}_t^{\mathbb{P}}[RV_{t,T}]. \tag{1}$$

For equity indices this difference is large and positive from the variance seller's point of view (Carr and Wu, 2009). It stems from a structural imbalance between the demand for protection,

concentrated in out-of-the-money puts, and the supply from those who absorb tail risk. Market makers who take the other side are short gamma and therefore require compensation for the losses they suffer when prices jump (Ang et al., 2018; Israelov, 2019).

Model-free implied volatility (MFIV) is the square root of the 30-day variance swap strike. The VIX methodology computes it from the out-of-the-money options of the two expirations around 30 days (Cboe Exchange, Inc., 2019):

$$\sigma^2_{\text{MFIV}} = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\, e^{rT}\, Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2, \tag{2}$$

where $Q(K_i)$ is the mid price of the out-of-the-money option with strike K_i, F is the index forward level and K_0 is the first strike below F. The VIX divided by 100 is, by construction, the 30-day MFIV of the S&P 500, and it is used as such in this paper.

2.2 Delta-hedged short volatility

A short straddle (selling a call and a put with the same strike and expiry) starts with delta close to zero and negative gamma. If the position is hedged continuously, its daily P&L is approximately

$$d\Pi_t \approx \tfrac{1}{2}\,\Gamma_t S_t^2 \left(\sigma^2_{\text{imp}} - \sigma^2_{\text{real},t}\right) dt, \tag{3}$$

with $\Gamma_t < 0$ for the short position (Sinclair, 2008). The seller gains when realised volatility stays below implied and loses otherwise, and the loss grows convexly with the size of the move. Systematic option-selling indices display this profile: from 1986 to 2018 the CBOE PutWrite index had a volatility of 9.9% against 14.9% for the S&P 500, but a skewness of -2.10 and a kurtosis of 9.72 (Wilshire Associates, 2019).

In practice hedging is discrete. Rebalancing often reduces gamma risk but multiplies costs; rebalancing rarely saves costs but leaves the position exposed. The hedge tolerance band is therefore a first-order economic parameter, as Section 5.5 confirms.

2.3 Machine learning and time-series validation

Gradient-boosting models approximate the target function with a sum of regression trees fitted sequentially to the pseudo-residuals of the previous ensemble (Friedman, 2001). XGBoost uses a second-order approximation of the loss and L1 and L2 regularisation of leaf weights (Chen and Guestrin, 2016); LightGBM grows trees leaf-wise and uses gradient-based sampling (Ke et al., 2017). Compared with linear models and the GARCH family (Bollerslev, 1986), they capture non-linearities and interactions without specifying them in advance, an advantage documented in forecasting problems with many predictors (Medeiros et al., 2021).

In financial time series, random cross-validation introduces look-ahead bias. When the target is built with future information, as here with the volatility of the next 21 sessions, training and test labels also overlap. López de Prado (2018) proposes purging the training observations whose labels overlap the test block and adding an embargo period.

3 Data

The sample runs from 2 January 2018 to 31 December 2025: 2,011 sessions and 96 monthly expirations (Table 1). SPX options are used rather than SPY options because they are European-style, cash-settled and form the deepest volatility market, on which the VIX itself is computed (Cboe Exchange, Inc., 2019).

Table 1: Data sources

Source	Series	Use
ThetaData (Options Standard plan)	SPX monthly options (third Friday): end-of-day quotes, greeks, implied volatility and open interest	Predictors, position selection and valuation
Stooq (yfinance as fallback)	SPX daily open, high, low and close	Realised volatility, hedging, buy & hold
FRED	VIXCLS, DGS3MO, DGS10	MFIV, rates, curve slope, collateral
yfinance	CBOE SKEW index	Tail-risk predictor

The chain is cleaned with five filters: positive volume and open interest of at least 100 contracts; valid quotes ($0 < \text{bid} < \text{ask}$); relative spread no greater than 20% of the mid price; 15 to 45 days to expiry; and moneyness (K/S) between 0.85 and 1.15. After cleaning, 341,094 observations remain, about 7–9% of the raw chain.

The 2021 chains could not be retrieved because of a download failure. For that year the model has the market and realised-volatility predictors but not the option microstructure ones, and the option strategy does not trade. The variance swap analysis (Section 5.7) does not depend on options and does cover 2021.

4 Methodology

4.1 Target

The target is the realised 21-session volatility premium:

$$y_t = \sigma_{\text{MFIV},t} - \sigma^{\text{YZ}}_{(t,\,t+21]}, \tag{4}$$

where $\sigma_{\text{MFIV},t}$ is the VIX divided by 100 and σ^{YZ} is the Yang and Zhang (2000) estimator over the next 21 sessions. This estimator combines the overnight jump, the open-to-close move and the Rogers–Satchell term, and is unbiased in the presence of drift and opening jumps. The premium is expressed in volatility: 0.05 equals 5 volatility points.

4.2 Predictors

The 22 predictors (Table 2) use only information available at the close of session t and fall into three families: realised volatility over 21-session windows, with the close-to-close, Parkinson (1980), Garman and Klass (1980) and Yang–Zhang estimators; option-market microstructure, measured on the expiry closest to 30 days; and market and momentum variables.

Table 2: Predictors

Family	Variables
Realised volatility (21 sessions)	`rv_cc`, `rv_parkinson`, `rv_gk`, `rv_yz`
Option microstructure	`atm_iv`; `skew_slope` (IV of the −0.25-delta put minus IV of the 0.25-delta call); `pcr_volume`, `pcr_oi`; `iv_rank`, `iv_percentile` (252 sessions)
Market and momentum	`mfiv`, `vix`, `vix_chg`, `skew`, `skew_chg`, `ust_3m`, `ust_10y`, `term_slope`; `ret_5d`, `ret_10d`, `ret_21d`, `ret_63d`

4.3 Models and validation

XGBoost and LightGBM are compared with a Huber loss (Huber, 1964) with threshold $\delta = 0.02$, which is quadratic for small errors and linear for large ones and so limits the influence of extreme episodes such as March 2020. Hyper-parameters are chosen by a random search of 40 combinations on a grid (Appendix A), with a fixed seed.

Validation is time-ordered with an expanding window and five folds. Between the end of each training block and the start of the test block, $21 + 5$ sessions are dropped: 21 of purge (the label length) and 5 of embargo. In the original thesis, the hyper-parameter search and the choice between XGBoost and LightGBM were scored on the same folds that were then reported. This revision adds **nested validation**: for each outer fold, both models run their search on a three-fold purged inner validation inside the training block, and the model and parameters with the lowest inner loss are refitted on the whole block and predict the test block. No decision evaluated on a block uses information from that block.

4.4 Naive benchmarks and tests of predictive accuracy

The model is compared with two forecasts without learning, scored on the same folds: the mean of the target in the training block, and the naive premium $\sigma_{\mathrm{MFIV},t} - \sigma^{\mathrm{YZ}}_{(t-21,\,t]}$, which assumes realised volatility stays at its recent level. Equal predictive accuracy is tested with the Diebold and Mariano (1995) statistic, using a Newey and West (1987) long-run variance with 20 lags (because the 21-session labels overlap) and the small-sample correction of Harvey et al. (1997).

4.5 Option strategy

The strategy is that of the original thesis. When no position is open and the session's signal exceeds the threshold, it sells an at-the-money straddle on the expiry closest to 30 days. The position is marked daily at chain mid prices and hedged with the index whenever the absolute net delta exceeds the band $b\times$ contracts $\times$ multiplier. Each hedge trade pays 1 basis point of slippage and each option 1 dollar of commission, and the position is settled at expiry at intrinsic value. When an option has no quote, which happens systematically in the last 15 days because of the maturity filter, it is valued with Black–Scholes (Black and Scholes, 1973; Merton, 1973) at the last observed implied volatility. Size is one contract on USD 1 million of capital; with the SPX at 2,943 at the start of the window, each leg is about USD 294,000 of notional.

The evaluation window runs from 29 April 2019, the first out-of-sample forecast, to 31 December 2025. On this basis two configurations are defined:

- **Thesis configuration**: entry at mid, no interest on collateral, Sharpe ratio with a fixed 2% risk-free rate, and band $b = 0.50$, chosen in the original thesis for its result on the same sample.
- **Rigorous configuration**: sales at the bid, collateral accruing daily at the three-month Treasury bill yield (DGS3MO), Sharpe computed on returns in excess of that same rate, predictions from the nested validation, and a band chosen in each fold. The band is the one that maximises the Sharpe of unconditional selling on the data before the test block.

To measure the value of the forecast's magnitude, not just its sign, five signals are compared: always sell; ML > 0 (the original rule); ML $>$ median, which sells only when the forecast exceeds the median realised premium of the training block; ML sized, which sells $\min(\max(\hat{y}_t/m, 0), 3)$ contracts, with m that same median; and naive $>$ median, which applies the same threshold to the naive premium. All thresholds are set with training-block information only.

4.6 Synthetic variance swaps

To measure the premium independently of the option engine, I simulate the sale of 21-session variance swaps struck at $K = \mathrm{VIX}_t$ (in volatility points). The P&L per unit of vega notional is

$$\pi_t = \frac{K_t^2 - RV_{(t,\,t+21]}}{2K_t}, \qquad RV_{(t,\,t+21]} = \frac{252}{21}\sum_{i=1}^{21} r_{t+i}^2 \times 10^4, \tag{5}$$

with r the daily SPX log returns. A swap with USD 100,000 of vega notional gains or loses USD 100,000 per point of π_t. The strategy sells one swap every 21 sessions, without overlap, and statistics are averaged over the 21 possible entry calendars. As a robustness check, realised volatility is capped at 2.5 times the strike, as is customary. The analysis ignores that swaps trade somewhat above VIX squared because of convexity and jumps.

4.7 Inference on strategies

Confidence intervals for the Sharpe ratio and for Sharpe differences between strategies are obtained with the stationary bootstrap of Politis and Romano (1994), with a mean block length of 21 sessions and 10,000 resamples. Both series in each comparison are resampled with the same blocks, in the spirit of Ledoit and Wolf (2008).

5 Results

5.1 The premium: stylised facts

Figure 1 shows implied volatility and the volatility realised over the next 21 sessions. The premium of Equation (4) is positive on 87.4% of sessions, with a mean of 0.0495 and a standard deviation of 0.0724. Its distribution has a pronounced left tail: the premium turns sharply negative when realised volatility accelerates faster than implied, as in January–February 2018 or February–March 2020, with a minimum of −0.486.

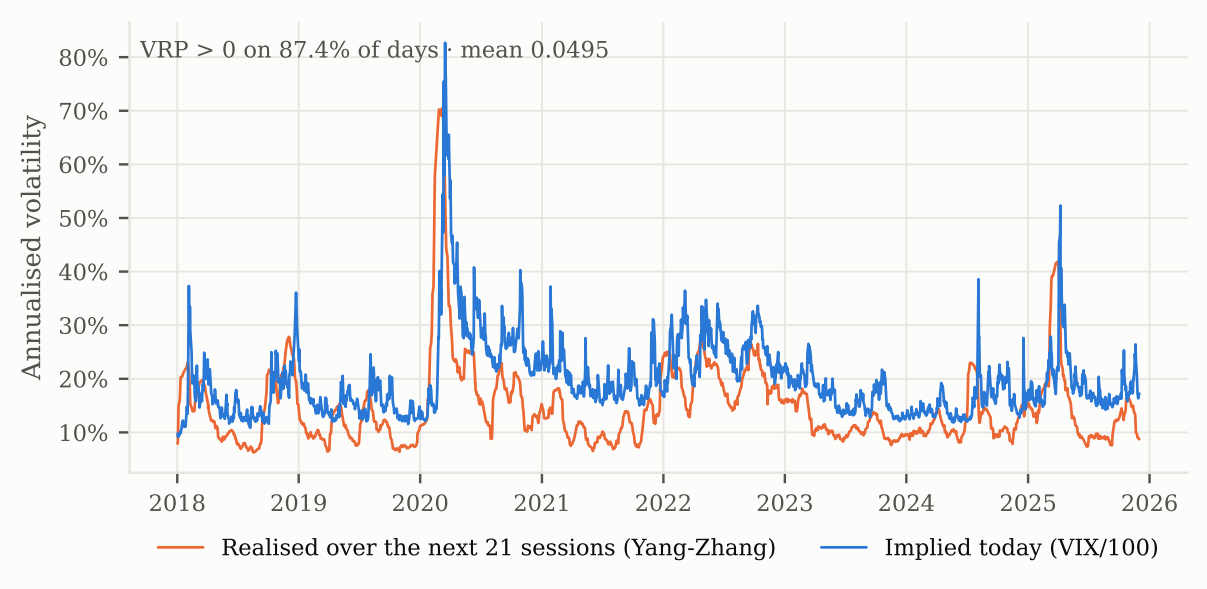


Figure 1: Implied volatility (VIX/100) and volatility realised over the next 21 sessions (Yang–Zhang), 2018–2025.

Table 3 and Figure 2 present the same premium as a variance swap sold every session. The crisis years (2018 and 2020) have a negative mean result, and 2021 is the best year in the sample:

the short swap earned 5.4 volatility points on average and was profitable on 95% of days. That the option strategy cannot trade in 2021 for lack of data works against it.

Table 3: Short 21-session variance swap, daily entry (volatility points per unit of vega notional)

Year	Mean K	Mean $\sqrt{RV}$	Mean P&L	% positive days	Worst day
2018	16.6	15.9	−1.0	61	−25.1
2019	15.4	11.2	3.1	84	−15.1
2020	29.3	27.1	−7.3	77	−252.2
2021	19.7	12.8	5.4	95	−2.2
2022	25.6	23.7	1.3	65	−19.8
2023	16.9	12.6	3.7	100	0.1
2024	15.6	12.4	2.4	81	−11.2
2025	19.3	16.0	0.8	83	−59.3

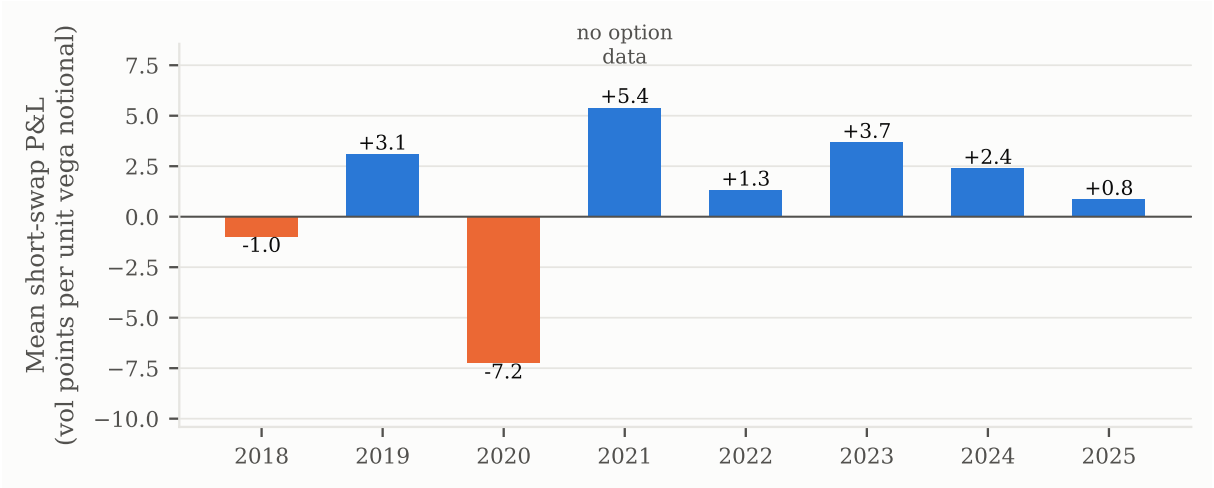


Figure 2: Mean annual result of a short 21-session variance swap with daily entry. 2021, the year without option chains, is the most favourable for the seller.

5.2 Forecast accuracy

Table 4 reports the out-of-sample metrics. XGBoost, the model selected in the original thesis, achieves a mean RMSE of 0.0662, 6% below the training mean (0.0704) and 14% below the naive premium (0.0766). The improvement is not uniform: the model loses to the training mean in folds 3 and 5 (Figure 3). Under nested validation, which selects LightGBM in the first fold, the mean RMSE rises to 0.0675 and the correlation falls from 0.21 to 0.16. Part of the original advantage therefore came from choosing hyper-parameters on the same evaluation data.

Table 4: Out-of-sample forecast accuracy: RMSE by fold and means

	RMSE by fold					Mean across folds			
Forecast	1	2	3	4	5	RMSE	MAE	Corr.	Dir. acc.
XGBoost (thesis)	0.1289	0.0437	0.0510	0.0356	0.0718	0.0662	0.0426	0.212	0.905
Nested	0.1329	0.0468	0.0506	0.0359	0.0716	0.0675	0.0473	0.156	0.898
Training mean	0.1295	0.0680	0.0463	0.0386	0.0698	0.0704	0.0491	–	0.905
Naive	0.1581	0.0490	0.0539	0.0371	0.0848	0.0766	0.0512	0.110	0.848

Test folds: Apr 2019–Aug 2020, Aug 2020–Dec 2021, Dec 2021–Apr 2023, Apr 2023–Jul 2024 and Aug 2024–Dec 2025.

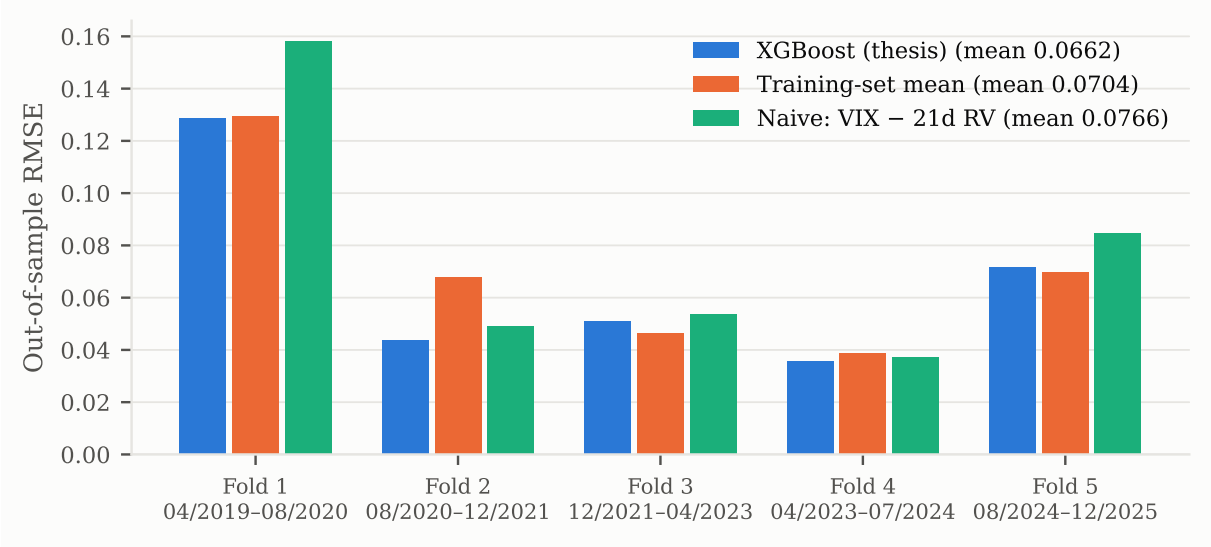


Figure 3: Out-of-sample RMSE by purged validation fold: XGBoost against the training mean and the naive premium.

The 90.5% directional accuracy highlighted in the original thesis carries no information. It equals that of a constant positive forecast, because the premium was positive on 90% of test days, and Figure 4 shows why: the XGBoost forecast ranges from 0.021 to 0.181 and is not negative in any of the 1,659 sessions. The Huber loss and the predominance of positive premia in training lead the model to forecast the typical level of the premium and not to anticipate its drops.

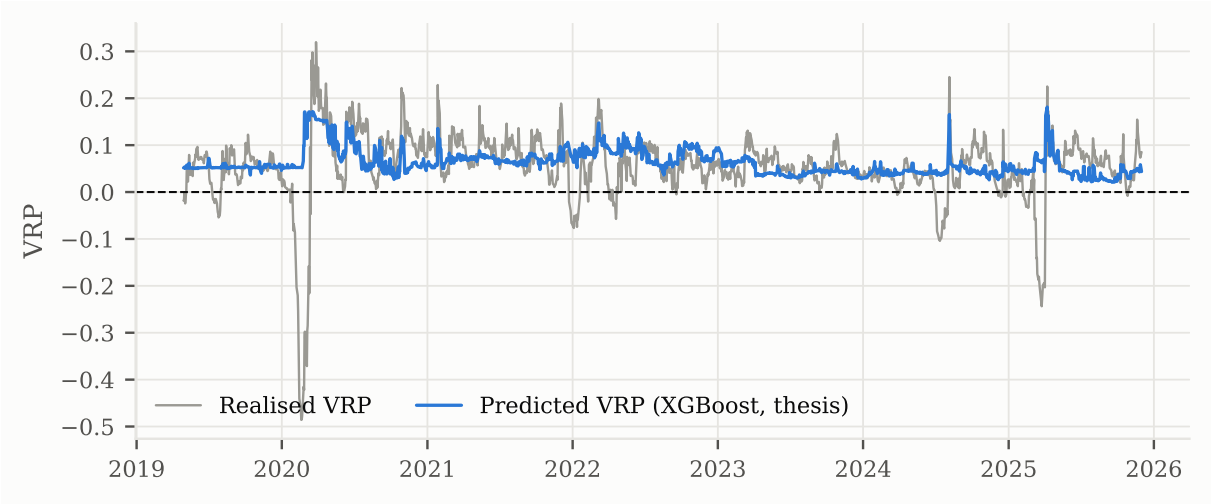


Figure 4: Premium forecast by XGBoost (thesis) and realised premium. The forecast never crosses zero.

Table 5 tests the difference formally. In squared error, the improvement of XGBoost over the training mean is not significant ($p = 0.27$); in absolute error it is ($p = 0.009$), which indicates that the model does better on normal days without improving on extreme days, which dominate squared error. Against the naive premium the evidence is weak (p between 0.03 and 0.06). Under nested validation no improvement is significant at 5%.

Feature importance of the thesis model, refitted on the full sample, is concentrated in the level of implied volatility (`mfiv` 0.126 and `vix` 0.123), followed by the three-month and ten-year yields (0.079 and 0.053) and at-the-money implied volatility (0.052). This is economically coherent, since the starting level of implied volatility is the main driver of the premium, but it is an in-sample measure and should not be read as causal evidence.

Table 5: Diebold–Mariano test (H_1: the model is more accurate). HLN-corrected statistic and one-sided p-value; $n = 1{,}659$

		Squared error		Absolute error	
Model	Benchmark	DM	p	DM	p
XGBoost (thesis)	Training mean	−0.61	0.272	−2.37	0.009
XGBoost (thesis)	Naive	−1.52	0.064	−1.91	0.028
Nested	Training mean	−0.87	0.192	−1.05	0.147
Nested	Naive	−1.63	0.052	−1.12	0.132

5.3 From forecast to trade

That the model forecasts something does not mean the information is useful for trading. To measure this directly, Figure 5 relates, for the 64 trades of unconditional selling, the outcome of each straddle to the forecast premium at entry. The Spearman rank correlation is 0.05 ($p = 0.33$ in a permutation test) with the thesis forecasts and −0.04 with the nested ones. The naive premium does not rank trades either (0.05), whereas the VIX level at entry does so weakly (0.21; $p = 0.05$). The distribution of outcomes illustrates the strategy's risk: the mean result is USD 929 and 59% of trades are winners, but the worst one, opened on 21 February 2020, loses USD 43,100, the equivalent of 46 average trades.

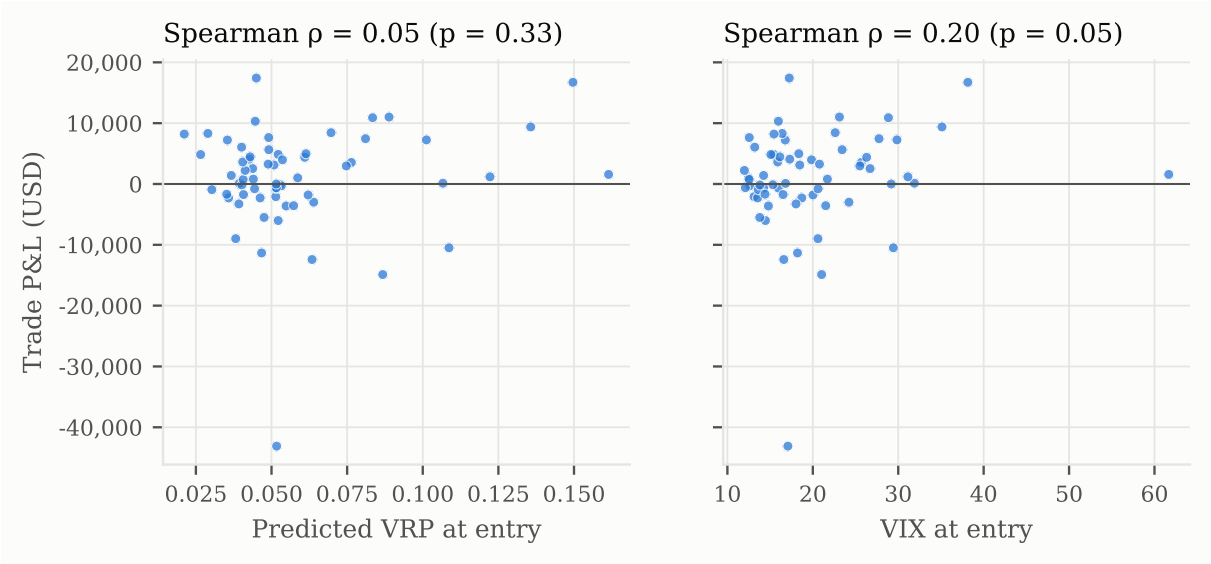


Figure 5: Outcome of each of the 64 trades of unconditional selling against the forecast premium and the VIX at entry.

5.4 The comparison with the systematic rule, revisited

The original thesis concluded that the model added tangible value over a systematic rule based on the naive premium: a cumulative return of 5.95% against 3.89%, with lower risk and greater selectivity (64 trades against 76). Table 6 reproduces those results exactly and shows why the conclusion does not hold.

The comparison was not like-for-like for two reasons. First, because every forecast is positive and the entry threshold is zero, the forecast-driven strategy sells whenever it is flat: its simulation matches that of always selling session by session (first and second columns), and the model takes part in no decision. Second, the systematic strategy was simulated from January 2018, because its signal is available from the start, and its equity curve was then cut to the evaluation window;

Table 6: The original comparison, revisited (thesis configuration, 29/04/2019–31/12/2025)

	ML (thesis)	Always sell	Systematic (from 2018)	Systematic (from window start)
Total return (%)	5.95	5.95	3.89	3.92
CAGR (%)	0.87	0.87	0.57	0.58
Annual volatility (%)	2.49	2.49	2.54	2.46
Sharpe (rf 2%)	−0.44	−0.44	−0.55	−0.57
Max drawdown (%)	−5.23	−5.23	−6.13	−5.93
Daily CVaR 95% (%)	0.45	0.45	0.46	0.44
Trades in window	64	64	62 (76)	63

In brackets, the figure in the original thesis, which included 14 trades opened between January 2018 and April 2019.

the trade count, however, was not. Within the window the systematic rule opens 62 trades, not 76, and it is the one that occasionally abstains, because its signal is negative on 6.4% of days. The return difference comes from those abstentions and from the rule entering the window with a position already open; started on the same day as the forecast-driven strategy, its result barely changes (3.92%). Figure 6 shows the three curves.

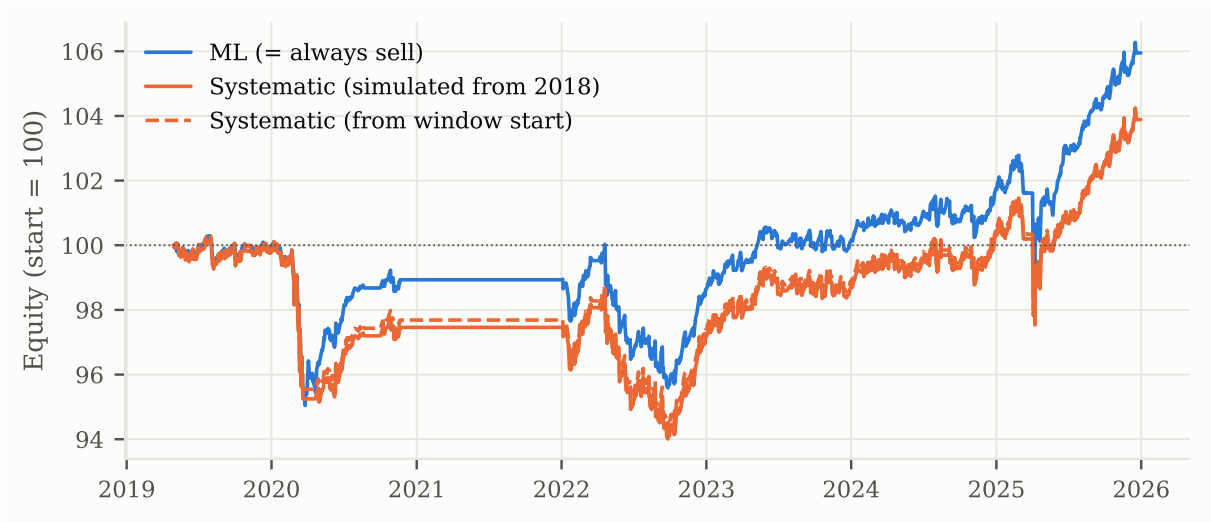


Figure 6: Equity of the forecast-driven strategy, identical to always selling, and of the systematic rule, simulated from 2018 or from the start of the evaluation window.

The correct conclusion is that, in this configuration, the model adds no value over the systematic rule or over unconditional selling. The original thesis did show that the premium exists and that hedging conditions the outcome; what it did not show is that the forecast improves trading.

5.5 Costs, collateral and hedging

The negative Sharpe ratio of the original thesis needs a qualification. In the simulation, the USD 1 million of capital earns no interest, yet the Sharpe ratio subtracts 2% a year. With a one-contract size, the strategy's return is small relative to that hurdle, and the sign of the Sharpe ratio reflects the convention more than the strategy. Table 7 and Figure 7 decompose the effect of each assumption on unconditional selling.

Paying the T-bill rate on collateral, as happens with the margin of a real portfolio, and measuring excess returns over that rate raises the Sharpe ratio from −0.44 to 0.33. Selling at the bid rather than at mid costs 1.7 points of cumulative return and lowers the Sharpe ratio to 0.23.

Table 7: Unconditional selling: cumulative effect of each assumption (29/04/2019–31/12/2025)

	Thesis (mid, rf 2%)	+ collateral at 3M T-bill	+ sell at bid	+ nested band
Total return (%)	5.95	26.08	24.33	11.62
CAGR (%)	0.87	3.54	3.32	1.66
Annual volatility (%)	2.49	2.28	2.30	3.66
Sharpe	−0.44	0.33	0.23	−0.29
Max drawdown (%)	−5.23	−4.61	−4.66	−10.45

From the second column on, the Sharpe ratio is computed on returns in excess of the three-month T-bill (2.8% on average over the window).

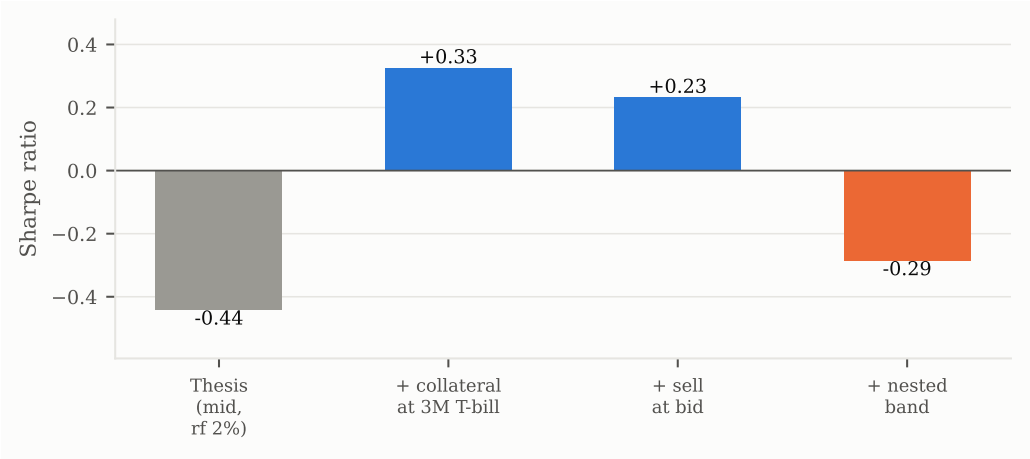


Figure 7: Sharpe ratio of unconditional selling as each assumption is added.

With correct accounting, delta-hedged short volatility earns a positive, if small, excess return.

The last step is the most important one. The 0.50 band was chosen in the original thesis as the one that maximised return on the same sample being evaluated (Figure 8): a narrow band eats the premium in costs (886 rebalances at 0.05) and no hedging leaves gamma exposed (maximum drawdown of −15.85%). When the band is chosen in each fold using prior data, the choice jumps around with no pattern: unhedged, 1.00, 0.05, 1.00 and 0.50 (Table 8). The Sharpe ratio drops to −0.29 and the maximum drawdown doubles. The inverted U of Figure 8 is real in sample, but its peak cannot be identified in advance.

Table 8: Nested band choice: Sharpe of unconditional selling on the data before each fold

Fold (data up to)	0.05	0.10	0.25	0.50	1.00	Unhedged	Chosen
1 (20/03/2019)	−1.54	−1.51	−1.72	−1.91	−1.41	**−1.12**	Unhedged
2 (13/07/2020)	−0.57	−0.75	−0.86	−0.81	**−0.52**	−0.75	1.00
3 (02/11/2021)	**−0.27**	−0.40	−0.48	−0.58	−0.36	−0.57	0.05
4 (28/02/2023)	−0.59	−0.63	−0.64	−0.39	**−0.36**	−0.54	1.00
5 (24/06/2024)	−0.30	−0.31	−0.35	**−0.27**	−0.46	−0.60	0.50

Every training period includes 2018 and, from fold 2 on, March 2020, hence the negative Sharpe ratios.

5.6 Does the forecast's magnitude help?

The original rule used only the sign of the forecast. Table 9 evaluates the alternatives that use its magnitude in the rigorous configuration, with the nested band and with the fixed 0.50 band, and Figure 9 shows the equity curves.

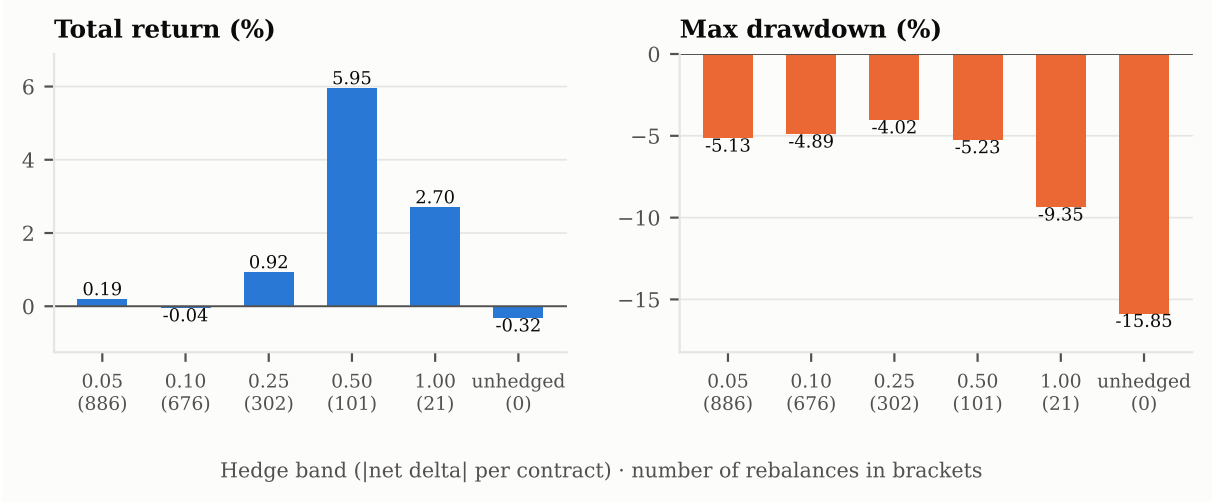


Figure 8: Sensitivity of total return and maximum drawdown to the hedge band (thesis configuration, forecast-driven strategy).

Table 9: Using the forecast's magnitude (rigorous configuration, 29/04/2019–31/12/2025)

	Always sell	ML > 0	ML > median	ML sized	Naive > median	Buy & hold
Band chosen in each fold (nested)						
Total return (%)	11.62	11.80	20.14	15.25	16.36	132.60
Annual volatility (%)	3.66	3.65	1.65	2.61	3.25	20.23
Sharpe	−0.29	−0.28	0.00	−0.23	−0.13	0.59
Max drawdown (%)	−10.45	−10.53	−2.44	−5.95	−10.35	−33.93
Daily CVaR 95% (%)	0.60	0.60	0.28	0.45	0.49	3.07
Band fixed at 0.50						
Total return (%)	24.33	24.50	23.12	23.39	25.19	–
Annual volatility (%)	2.30	2.30	1.69	2.12	1.81	–
Sharpe	0.23	0.24	0.22	0.20	0.35	–
Max drawdown (%)	−4.66	−4.65	−3.29	−5.26	−4.33	–
Daily CVaR 95% (%)	0.40	0.40	0.28	0.38	0.30	–
Trades	64	64	30	64	42	–

ML sized: 0.82 contracts on average. Sharpe is the excess over the three-month T-bill; collateral earns that rate.

With the nested band, trading only when the forecast premium exceeds the historical median halves volatility, cuts the maximum drawdown from −10.5% to −2.4%, and raises the Sharpe ratio from −0.29 to 0.00. The result looks attractive at first sight, but it comes mainly from trading less (30 trades against 64) in a configuration where the hedge is badly calibrated: with the band fixed at 0.50, the same rule does not beat unconditional selling (0.22 against 0.23). Proportional sizing does not help in either configuration. The naive rule with the same threshold obtains the best Sharpe with the fixed band (0.35), so if filtering on the level of the premium has any value, it does not require a machine-learning model.

5.7 The premium in pure form: variance swaps

Table 10 presents the results of selling monthly variance swaps, averaged over the 21 possible entry calendars, and Figure 10 shows the cumulative result and the dispersion across calendars.

Variance swaps confirm the conclusions of the option strategy in a setting that does not

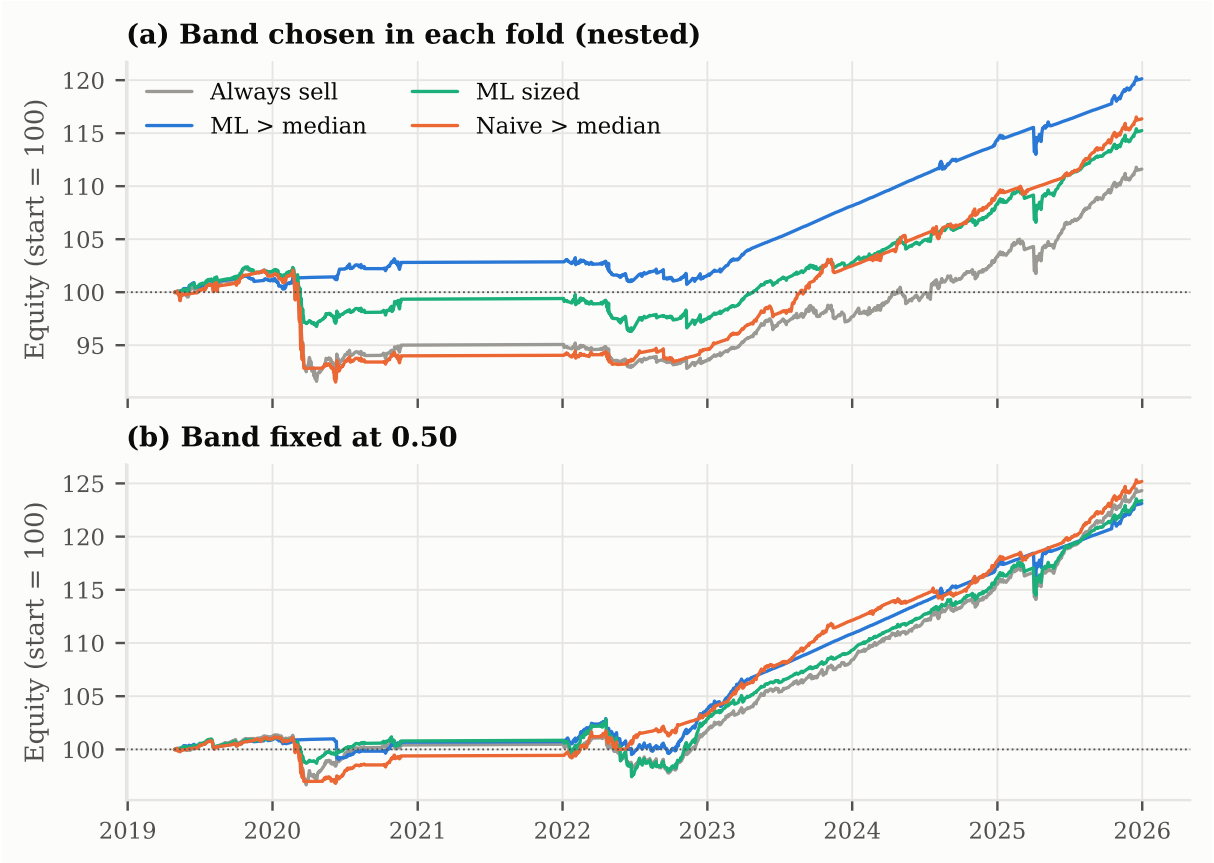


Figure 9: Equity of the strategies in the rigorous configuration (sales at the bid, interest-bearing collateral), with the band chosen in each fold and with the band fixed at 0.50.

Table 10: Variance swaps sold every 21 sessions (mean over the 21 entry calendars; P&L in volatility points per unit of vega notional)

Strategy	Months with position	Uncapped Mean P&L	Uncapped Sharpe	Uncapped Skewness	Uncapped Worst month	Capped at 2.5·K Mean P&L	Capped at 2.5·K Sharpe
Unconditional, 2018–2025	94.8	1.06	0.28	−6.9	−145.8	1.97	0.71
Unconditional, window	79.0	1.22	0.31	−6.8	−145.8	2.32	0.81
ML > 0, window	78.4	1.19	0.31	−6.7	−145.8	2.29	0.80
ML > median, window	36.0	0.31	0.50	−3.5	−96.1	1.13	0.70
Naive > median, window	38.6	1.03	0.71	−3.9	−77.3	1.32	0.81

Mean P&L includes months without a position (zero result). Annualised Sharpe of monthly results. Window: 29/04/2019–31/12/2025.

depend on execution, hedging or option data. The unconditional premium is positive in 81–83% of months, but its distribution is very negatively skewed (−6.9), and without a cap a single month of 2020 wipes out years of results. Filtering with the model's forecast improves the mean uncapped Sharpe (0.50 against 0.31) at the cost of trading in fewer than half the months, but not the capped one (0.70 against 0.81). The naive rule again does as well or better. Moreover, the dispersion across calendars is very large: depending on the entry date, the Sharpe of the model-filtered strategy ranges from −0.25 to 2.05. With about 80 months, the entry date matters as much as the signal.

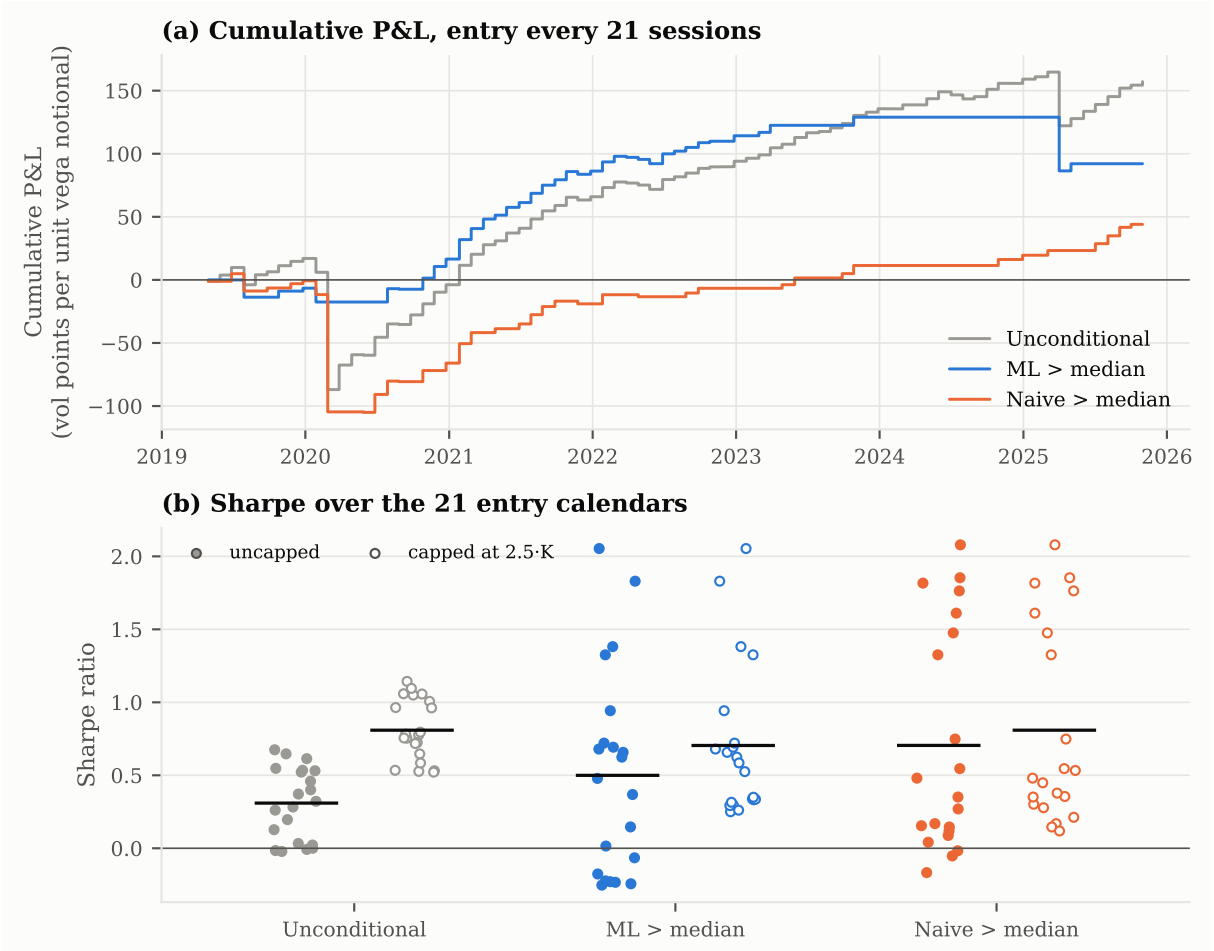


Figure 10: Variance swaps sold every 21 sessions: (a) cumulative result with the first entry calendar; (b) Sharpe ratio of each of the 21 calendars, uncapped and capped at $2.5 \cdot K$ (black lines are the means).

5.8 Volmageddon

Figure 11 places the February 2018 episode in this framework. This part is descriptive, because the option backtest starts in 2019. Throughout January 2018 the VIX stayed between 9 and 15, while for swaps sold from 9 January onwards the volatility realised over the next 21 sessions was 23.5 to 26.4 points. A swap sold on 22 January with a strike of 11.0 realised 24.8 points and lost 22.3 points per unit of vega notional, that is, USD 2.2 million per USD 100,000 of vega notional, the equivalent of about 20 months of average premium (1.06 points a month over 2018–2025). The loss did not come from the session of 5 February itself, but from having sold volatility when implied was historically low. Once the jump had occurred, implied exceeded subsequently realised volatility, and swaps sold between 5 and 21 February earned between 3.2 and 12.9 points.

5.9 Inference

Figure 12 and Table 11 summarise the bootstrap confidence intervals. No Sharpe ratio of the volatility strategies differs from zero at 95% confidence, and no difference against unconditional selling is significant: the largest, ML > median with the nested band, has a 95% interval of $[-0.45, 0.95]$. The available sample cannot tell the variants apart, and any ranking in Table 9 should be read with that caution.

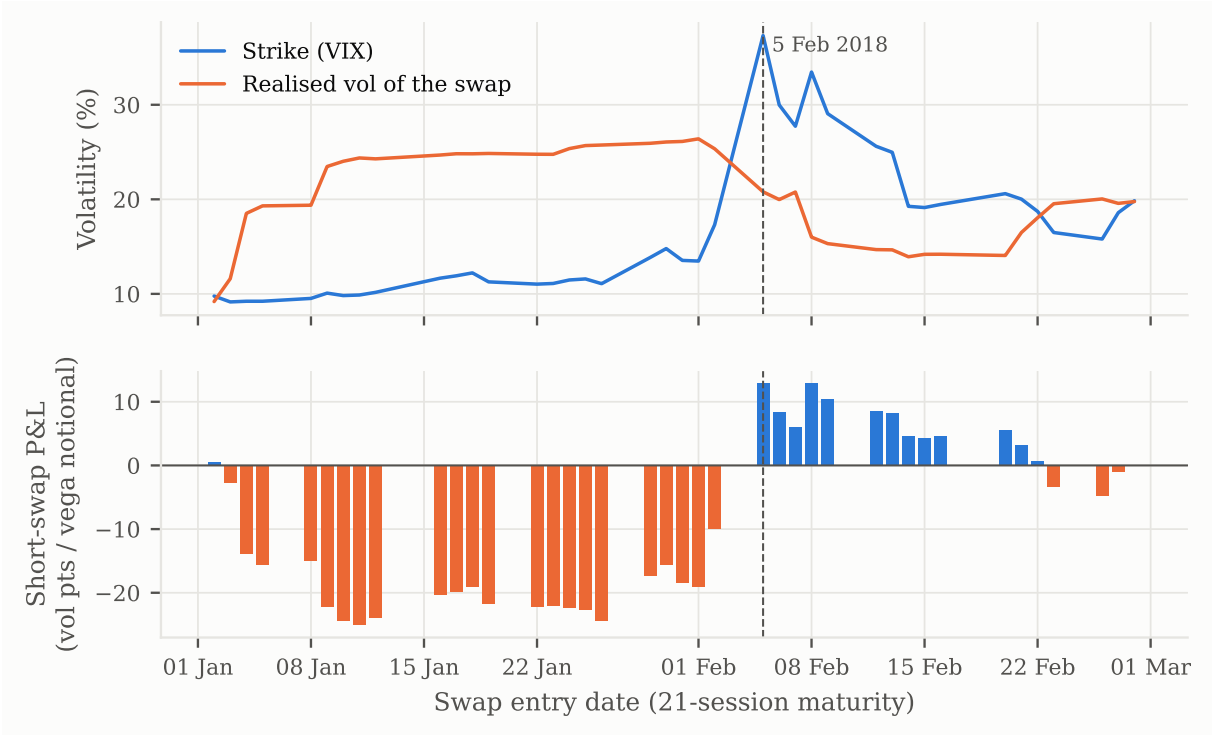


Figure 11: Variance swaps sold in January and February 2018: strike (VIX) and volatility realised over the next 21 sessions (top), and result by entry date (bottom). Descriptive analysis; not part of the option backtest.

Table 11: Stationary-bootstrap confidence intervals (rigorous configuration, nested band; 10,000 resamples)

	Estimate	95% CI	90% CI	P(≤ 0)
Sharpe: always sell	−0.29	[−0.84, 0.49]	[−0.76, 0.38]	0.76
Sharpe: ML > median	0.00	[−0.55, 0.60]	[−0.46, 0.50]	0.49
Sharpe: buy & hold	0.59	[−0.03, 1.35]	[0.07, 1.23]	0.03
Diff.: ML > 0 − always sell	0.01	[−0.02, 0.04]	[−0.01, 0.03]	0.37
Diff.: ML > median − always sell	0.29	[−0.45, 0.95]	[−0.35, 0.85]	0.26
Diff.: ML sized − always sell	0.06	[−0.21, 0.24]	[−0.16, 0.21]	0.36
Diff.: naive > median − always sell	0.15	[−0.18, 0.72]	[−0.12, 0.61]	0.18
Swap diff.: ML > median − uncond.	0.09	[−1.62, 1.45]	[−1.43, 1.25]	0.53

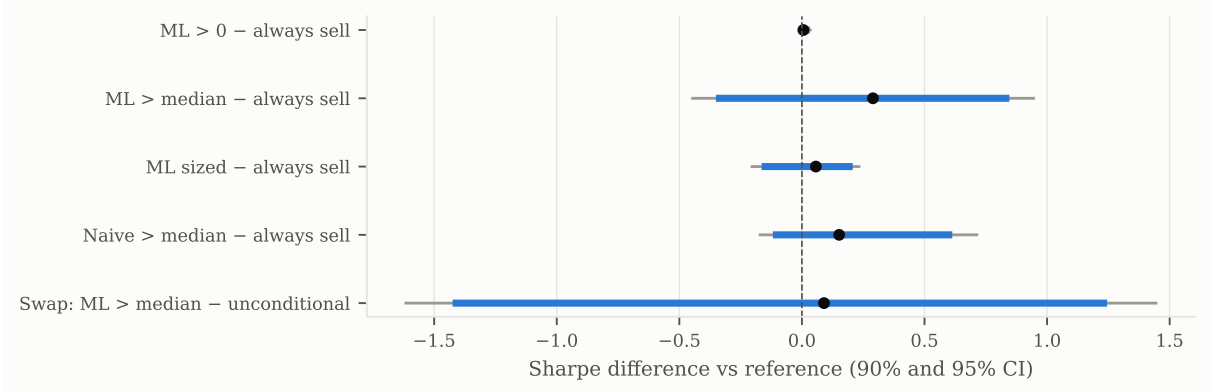


Figure 12: Sharpe differences against the reference, with 90% (thick bar) and 95% (thin bar) confidence intervals.

6 Limitations and extensions

Valuation at the end of the option's life. The cleaned chain covers 15 to 45 days to expiry, so the last 15 days of each straddle, when gamma peaks, are valued with Black–Scholes at the last implied volatility. This is the most important limitation for tail risk, and the only one that requires new option data: a tracking chain from 0 to 45 days would allow the whole life of the position to be marked at market prices.

The year 2021. The missing 2021 chains remove the most favourable year for the volatility seller (Table 3). The variance swap analysis covers it, but the option strategy does not.

Execution assumptions. Selling at the bid corrects the entry spread, but hedging is modelled with 1 basis point of slippage and no market impact, settlement uses the close on expiry day rather than the special opening quotation of monthly SPX options, and size is fixed.

Statistical power. The results rest on a single 6.7-year path, 64 trades and about 80 months, dominated by a bull market. The intervals in Section 5.9 show that this sample cannot distinguish between the variants.

Model selection. Nested validation removes the selection bias in hyper-parameters and in the band, but the overall design (predictors, horizon, loss, grid) was built with knowledge of the sample.

Extensions. The most promising lines are: (i) a complete tracking chain and the recovery of 2021; (ii) forecasting the hedged trade's outcome directly, instead of the volatility spread, or recasting the problem as classifying regimes of negative premium; (iii) sizing towards a volatility target using the forecast of risk rather than of the premium; (iv) extending the sample backwards and adding weekly options (SPXW) to increase the number of independent trades; and (v) testing the rules on other assets (VIX, VIX futures, other indices).

7 Conclusions

The S&P 500 variance risk premium is persistent and can be harvested: with correct accounting for collateral and sales at the bid, systematic selling of hedged straddles earns a positive excess return with a fraction of the risk of equities. Forecasting the premium with gradient boosting improves on the naive benchmarks on average, but modestly, unstably across periods and not significantly in squared error. That information does not carry over to trading: the model forecasts a positive premium in every session, its forecast does not rank trade outcomes, and rules that use its magnitude do not significantly improve on unconditional selling or on a naive rule.

The economic result is determined by risk management. The hedge band explains more of the variation in outcomes than any signal, and its in-sample optimum cannot be identified in advance. For a volatility seller, the relevant question is not how large the premium will be next month, since its typical level is well known, but when it may turn sharply negative and what it costs to protect against that. That is the problem a next version of this system should address.

References

I.-C. Ang, R. Israelov, R. Sullivan, and H. Tummala. Understanding the volatility risk premium. Technical report, AQR Capital Management, 2018.

Patrick Augustin, Ing-Haw Cheng, and Ludovic Van den Bergen. Volmageddon and the failure of short volatility products. *Financial Analysts Journal*, 77(3):35–51, 2021.

Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

Peter Carr and Liuren Wu. Variance risk premiums. *The Review of Financial Studies*, 22(3): 1311–1341, 2009.

Cboe Exchange, Inc. Cboe VIX white paper: Cboe volatility index. Technical report, Cboe Exchange, Inc., 2019.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001.

Mark B. Garman and Michael J. Klass. On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1):67–78, 1980.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

Peter J. Huber. Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–101, 1964.

Roni Israelov. Pathetic protection: The elusive benefits of protective puts. *The Journal of Alternative Investments*, 2019.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, pages 3146–3154, 2017.

Olivier Ledoit and Michael Wolf. Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5):850–859, 2008.

Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.

Marcelo C. Medeiros, Gabriel F. R. Vasconcelos, Álvaro Veiga, and Eduardo Zilberman. Forecasting inflation in a data-rich environment: The benefits of machine learning methods. *Journal of Business & Economic Statistics*, 39(1):98–119, 2021.

Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Michael Parkinson. The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1):61–65, 1980.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

Euan Sinclair. *Volatility Trading*. John Wiley & Sons, Hoboken, NJ, 2008.

C. Varadarajan and K. R. Schenk-Hoppé. BeVIXed: Trading fear in the volatility complex. *Risks*, 11(5):86, 2023.

Wilshire Associates. Options-based benchmark indexes: Performance, risk and premium capture (june 1986–dec. 2018): An update. Technical report, Cboe, 2019.

Dennis Yang and Qiang Zhang. Drift-independent volatility estimation based on high, low, open, and close prices. *The Journal of Business*, 73(3):477–492, 2000.

A Hyper-parameters

Table 12: Search grid (40 random combinations with seed 42) and choice by fold

Parameter	Values
Learning rate	0.01; 0.05; 0.1
Leaves (LightGBM)	31; 63; 127
Depth (XGBoost)	3; 5; 7
Row and column subsampling	0.7; 0.8; 0.9
L1 and L2 regularisation	0; 0.1; 1; 10
Trees; Huber threshold	400; $\delta = 0.02$
Thesis (XGBoost)	rate 0.1; depth 3; subsampling 0.9 and 0.9; L1 1; L2 0.1
Nested, fold 1	LightGBM: rate 0.1; 63 leaves; 0.8 and 0.8; L1 1; L2 10
Nested, folds 2–5	XGBoost: depth 3, 5, 7 and 3; otherwise as in the thesis except L2 1 (fold 3) and L1 0.1 (fold 4)

B Reproducibility

The code, organised as a Python package, is available at https://github.com/borja-izquierdo/vrp-sp500-options. The script `scripts/run_pipeline.py` reproduces the results of the original thesis and `scripts/run_extensions.py` those of this revision. ThetaData data cannot be redistributed; the repository includes the download script, the out-of-sample forecasts of the original thesis and every results table. Seeds are fixed (42). XGBoost does not produce bit-identical forecasts across platforms: retraining on Linux changes the fold RMSE by up to 0.003 relative to the original run, although every forecast remains positive and the backtest does not change.